Week 5 R Exercises

Question 1: A simple Gibbs sampler example

Suppose that we need to sample from the following joint distribution

$$f(x, y|\alpha, \beta) \propto \binom{n}{x} y^{x+\alpha-1}(1-y)^{n-x+\beta-1} \tag{1}$$

where $x = 0, 1, \ldots, n$, $0 \leq y \leq 1$ with $\alpha, \beta > 0$.

From the above equation,

$$\pi(x|y) \propto \binom{n}{x} y^{x}(1-y)^{n-x}$$

such that

$$X|Y \sim \text{Bin}(n, y).$$

Similarly,

$$Y|X \sim \text{Beta}(x+\alpha, n-x+\beta).$$

Posterior samples can be obtained from $f(x, y|\alpha, \beta)$ by sampling from $X|Y$ and $Y|X$ in turn.

Let $n = 16$, $\alpha = 2$ and $\beta = 10$. The following code can now be used to sample from $f(x, y|\alpha, \beta)$. Note that only the samples for X are retained here.

1. Change the following code so as to obtain the marginal posterior distribution of y.
2. Thereafter run the code three times (called chains). Store the three sets of results.
3. Plot the traceplots related to $[y]$ on the same figure. What do you notice?
4. Discard the first half of each of the samples. Now store these values in one vector named 'y.samples'. Why does it make sense to do this?
5. Plot the histogram and autocorrelation function of 'the'y.samples'. Comment on your results.

```
sample_x2<-function(k=10000, start.x=sample(1:16, size=1), start.y=runif(1)){
  #k = the length of a Gibbs sequence!
  #start.x and start.y are the starting values of x and y
  #these are random values

  n<-16; alpha<-2; beta<-10

  #starting valkue of y
  x<-NULL; y<-NULL
  y[1]<-start.y; x[1]<-start.x #starting values

  for (i in 2:k){
    x[i]<-rbinom(n=1, size=n, prob=y[i-1])
    y[i]<-rbeta(n=1, shape1= (x[i]+alpha), shape2= (n-x[i]+beta))
  }
  return(x)
}
```

Question 2: The Bayesian Linear model

Simulate data some regression data.

```
#Lets simulate some data
set.seed(1)
n <- 150
X.c <- data.frame(matrix(rnorm(5*n), ncol=5))
colnames(X.c) <- c("X1", "X2", "X3", "X4", "X5")

X <- as.matrix(cbind(1, X.c))
e <- matrix(rnorm(n), ncol=1)

beta.true <- matrix(c(1, 0, 10, 0, 2, -3), ncol=1)
Y <- X%*%beta.true + e
```

The OLS results

```
fit1 <-lm(Y~ X1 + X2 + X3 + X4 + X5, data=X.c)
summary(fit1)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X3 + X4 + X5, data = X.c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.65555 -0.75505  0.00037  0.79276  2.99348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.993211   0.089175  11.138   <2e-16 ***
## X1          -0.023844   0.098598  -0.242    0.809
## X2          10.052046   0.086969 115.582   <2e-16 ***
## X3           0.002165   0.086479   0.025    0.980
## X4           2.101651   0.083887  25.053   <2e-16 ***
## X5          -3.015520   0.080422 -37.496   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.084 on 144 degrees of freedom
## Multiple R-squared:  0.9911, Adjusted R-squared:  0.9908
## F-statistic:  3194 on 5 and 144 DF,  p-value: < 2.2e-16
```

Bayesian fit of the full model

Assume that the design matrix $\boldsymbol{X}$ is an $n \times (k+1)$ matrix, such that there are 2^k possible regression models to fit to the data (an intercept is included in all models). Define $\boldsymbol{\gamma} \in \Gamma = \{0,1\}^k$ as a binary vector which indicates which model is being fit to the data. e.g. if $\boldsymbol{\gamma} = c(1,0,1,\ldots,0)$ only X_1 and X_3 are included in the model. The associated design matrix is $\boldsymbol{X_\gamma}$ such that the fitted model is $\boldsymbol{Y} = \boldsymbol{X_\gamma \beta_\gamma} + \boldsymbol{e}$.

Lets assume that $\boldsymbol{\gamma} = c(1,1,1,1,1)$ such that we include all covariates in the regression model. Assume that the following prior distributions are specified:

$$\boldsymbol{\beta}_{\boldsymbol{\gamma}}|c,\sigma^2,\boldsymbol{X} \sim N(\mathbf{0}, c\sigma^2(\boldsymbol{X}_{\boldsymbol{\gamma}}^T\boldsymbol{X}_{\boldsymbol{\gamma}})^{-1}) \tag{2}$$

$$\pi(\sigma^2) \propto \frac{1}{\sigma^2} \tag{3}$$

$$\pi(c) = \frac{1}{c}I_{N^+}(c) \tag{4}$$

$$\boldsymbol{\gamma} = 0.5^k. \tag{5}$$

Above we use the G-prior to specify the conditional prior distribution for $\boldsymbol{\beta}_{\boldsymbol{\gamma}}$ while we specify an uninformative prior distribution for σ^2. The prior distribution for c is assumed to be an vague prior that assigns more weight to smaller values of c. Note that $c = 1, 2, 3, \ldots$. The $\boldsymbol{\gamma}$ vector is a vector of indicator values. If $\gamma_i = 1$ the ith explanatory variable is assumed to be included in the regression model while if $\gamma_i = 0$ the variable is not included in the model. A-priori we thus assign a 50% chance that any particular variable is included in the model and then assume independence to obtain the prior distribution for $\boldsymbol{\gamma}$.

During the course of the semester were able to derive the marginal distribution of $\boldsymbol{\beta}|\boldsymbol{y}$. In many real world applications we are unable to analytically work this out and will have to rely on some other techniques. Below we use the Gibbs sampler in order to obtain the marginal distribution of $\boldsymbol{\beta}|\boldsymbol{y}$.

The following conditional posteriors distributions can easily be obtained:

$$\boldsymbol{\beta}_{\boldsymbol{\gamma}}|c,\sigma^2,\boldsymbol{y} \sim N\left(\frac{c}{1+c}\hat{\boldsymbol{\beta}},\ \frac{\sigma^2 c}{1+c}(\boldsymbol{X}_{\boldsymbol{\gamma}}^T\boldsymbol{X}_{\boldsymbol{\gamma}})^{-1}\right) \tag{6}$$

$$\sigma^2|\boldsymbol{\beta}_{\boldsymbol{\gamma}},c,\boldsymbol{y} \sim IG\left(\frac{n+v_k+1}{2},\ \frac{1}{2}\left((\boldsymbol{y}-\boldsymbol{X}_{\boldsymbol{\gamma}}\boldsymbol{\beta}_{\boldsymbol{\gamma}})^T(\boldsymbol{y}-\boldsymbol{X}_{\boldsymbol{\gamma}}\boldsymbol{\beta}_{\boldsymbol{\gamma}}) + \frac{1}{c}\boldsymbol{\beta}_{\boldsymbol{\gamma}}^T\boldsymbol{X}_{\boldsymbol{\gamma}}^T\boldsymbol{X}_{\boldsymbol{\gamma}}\boldsymbol{\beta}_{\boldsymbol{\gamma}}\right)\right) \tag{7}$$

$$[c|\boldsymbol{\beta}_{\boldsymbol{\gamma}},\sigma^2,\boldsymbol{y}] = \tag{8}$$

where $R_{\boldsymbol{\gamma}} = \frac{1}{\sigma^2}\boldsymbol{\beta}_{\boldsymbol{\gamma}}^T\boldsymbol{X}_{\boldsymbol{\gamma}}^T\boldsymbol{X}_{\boldsymbol{\gamma}}\boldsymbol{\beta}_{\boldsymbol{\gamma}}$ and v_k is the number of explanatory variables included in the model.

1. Provide an expression for $[c|\boldsymbol{\beta}_{\boldsymbol{\gamma}},\sigma^2,\boldsymbol{y}]$.
2. Explain how one can sample from $[c|\boldsymbol{\beta}_{\boldsymbol{\gamma}},\sigma^2,\boldsymbol{y}]$.
3. Use the above conditional posterior distributions in order to obtain posterior samples (10 000) of $[\boldsymbol{\beta}_{\boldsymbol{\gamma}},\sigma^2,c|\boldsymbol{y}]$. Use the code below as a guide to answer the following questions. You will have to complete the code.
   (a) A trace plot is a time series plot of the retained MCMC samples. Plot the trace plots for c, σ^2 as well as those for some of the regression coefficients. What do you notice?
   (b) Plot the histograms of the retained MCMC samples for all of the parameters of the model. Comment on the 95% credibility interval. Are the results similar to those obtained using least squares to analyse the data?
   (c) Compare the marginal posterior mean for the β coefficients to those obtained analytically. Do they agree?
   (d) What is the posterior median for c?
   (e) Have a look at the autocorrelation function of the retained samples. What do you notice?

```r
#the number of iterations
nsims <- 20000
#discard half of the iterations
keep.index <- floor(nsims/2):nsims
k <- 5 #the number of covariates

#used for sampling from a multivariate normal
#you can use other packages if you like
require(MASS)

#the ols estimate of beta
b.ols <- solve(crossprod(X))%*%t(X)%*%Y
C.lam <- solve(crossprod(X))

#truncate the infinite sum to 10000
c.index <- 1:10000

#set some starting values c, sigma2
#any random allowable number
c. <- 5
sigma2 <- 1

#a constant
const1 <- -(k+3)/2

#create objects to store the posterior samples
beta.sims <- matrix(0, nrow=nsims, ncol=k+1)
sigma2.sims <- matrix(sigma2, nrow = nsims)
c.sims <- matrix(c., nrow = nsims)

#put the sampling code in a for loop
#start at isims=2
for (isims in 2:nsims){
  #sampling can be rather slow at times!

  #simulate 'beta' and store it in beta.sim
  #use the 'mvrnorm' function in the MASS package
  beta.sims[isims, ] <-

  #simulate 'sigma2' and store it in sigma2
  #have a look at the way R specifies 'rgamma'
  sigma2[isims] <-

  #simulate the value of 'c'
  #think about how to do this!!!
  #truncate the infinite sum to 10000
  R.lam <- (crossprod(X%*%beta.sims[isims, ])/sigma2[isims])[1]

  #add in a few lines here

  c.sims[isims] <-
}
```